MILITARY SPECIFICATION

CONTAINER, PALLET (PALCON)

This specification is approved for use within the United States Marine Corps, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification establishes the requirements for the manufacture and acceptance of the pallet container (PALCON). A container rack which provides support for six insert containers is required when the PALCON is used in its optional bin/drawer configuration.

2. APPLICABLE DODUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. The following specifications, standards and handbooks form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation:

SPECIFICATIONS

MILITARY

MIL-T-704 Treatment and Painting of Materiel

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commandant of the Marine Corps, Code LMA-2, Washington, D.C. 20380, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 8145

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MIL-C-5541	Chemical Conversion Coatings on Aluminum Alloys
MIL-Q-9858	Quality Program Requirements
MIL-C-29446(MC)	Container, Insert (INSERT)
MIL-C-29447(MC)	Container Rack, Pallet (PALCON Rack)

STANDARDS

MILITARY

MIL-STD-129	Marking for Shipment and Storage
MIL-STD-130	Identification Marking of U.S. Military Property
MIL-STD-209	Slinging and Tiedown Provisions for Lifting and Tying Down Military Equipment
MIL-STD-810	Environmental Test Methods and Engineering Guidelines
MIL-STD-831	Test Reports, Preparation of
MIL-STD-889	Dissimilar Metals
MIL-STD-1186	Cushioning, Anchoring, Bracing, Blocking and Waterproofing, with Appropriate Test Methods
MIL-STD-1791	Designing for Internal Aerial Delivery in Fixed Wing Aircraft

2.1.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this specification to the extent specified herein. Unless otherwise specified, the issues shall be those in effect on the date of the solicitation.

DRAWING

MARINE CORPS

835039A0000	Container, Pallet (PALCON)
835041A0000	Container Rack, Pallet (PALCON Rack)

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835043A0000 Container, Insert (INSERT)

(Copies of specifications, standards, handbooks, drawings, and publications required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting activity.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of the documents which are DOD adopted shall be those listed in the issue of the DODISS specified in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS shall be in the issue of the nongovernment documents which is current on the date of the solicitation.

AMERICAN SOCIETY FOR TESTING AND MATERIALS

ASTM-E-162	Surface Flammability of Materials Using a Radiant Heat Energy Source
ASTM-D-229	Rigid Sheet and Plate Materials used for Electrical Insulation
ASTM-D-256	Impact Resistance of Plastic and Electrical Insulating Materials
ASTM-D-570	Water Absorption of Plastics
ASTM-D-638	Tensile Properties of Plastics
ASTM-D-648	Deflection Temperature of Plastics Under Flexural Load
ASTM-D-695	Compressive Properties of Rigid Plastics
ASTM-D-790	Flexural Properties of Unreinforced and Reinforced Plastics and Electrical Insulating Materials
ASTM-D-792	Specific Gravity and Density of Plastics by Displacement
ASTM-D-1043	Stiffness Properties of Plastics as a Function of Temperature by Means of a Torsion Test
ASTM-D-2583	Indentation Hardness of Rigid Plastics by Means of a Barcol Impressor

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, Pennsylvania 19103.)

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U.S. DEPARTMENT OF COMMERCE

PS-1 Softwood Plywood, Construction and Industrial

(Application for copies should be addressed to the Superintendent of Documents, Government Printing Office, Washington, D.C. 20402.)

(Industry association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

2.3 Order of precedence. In the event of a conflict between the test of this specification and the references cited herein (except for associated detail specifications, specification sheets, or MS standards), the text of this specification shall take precedence. Nothing in this specification, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 First article. When specified in the contract or purchase order a sample shall be subjected to first article inspection (see 4.4 and 6.3). The first articles shall be manufactured using the same procedures and tooling proposed for production. Approval of the first articles by the acquiring activity shall not relieve the Contractor of the obligation to supply equipment conforming to this specification. Changes or deviations in production units from the approved first article baseline shall require written authorization by the acquiring activity.

3.2 Materials. Materials and components shall conform to the requirements specified herein and in Drawing Numbers (No.) 835039A0000, 835041A0000, and 835043A0000.

3.2.1 Plywood. The plywood core material shall conform to Department of Commerce PS-1, grade A-A (plugged) or better, bonded with exterior glue. All plywood for the floor shall be treated in accordance with Drawing No. 835039A0000 to prevent moisture absorption and rotting.

3.2.2 Fiberglass reinforced plastic. All container panels; side and back walls, doors, and roof, shall be fabricated from fiberglass reinforced plastic commonly known as sheet molding compound (SMC). The SMC sheet shall be formed by chopping continuous fiberglass rovings into 2-in lengths and blending with a thermosetting resin. The composite sheet shall contain not less than 42 percent glass and a maximum of 58 percent polyester resin by weight. The chopped glass rovings shall be randomly deposited to achieve structural uniformity. The exterior panel surfaces shall be ultraviolet; weather, mildew, oil and chemical resistant; and shall conform to the color and finish requirements of 3.9. The physical properties of the SMC shall meet the requirements specified in Table I.

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Table I. Physical Properties of SMC

PROPERTIES	TEST METHOD	VALUES
Impact, izod notched, ft/lb/in	ASTM-D-256	15.0
Impact, izod at room temperature	ASTM-D-256	12.0
Flexural strength psi	ASTM-D-790	20,000
Flexural modulus x 10^6 psi	ASTM-D-790	1.50
Tensile strength psi	ASTM-D-638	12,000
Tensile modulus x 10^6 psi	ASTM-D-638	1.50
Compressive strength psi	ASTM-D-695	28,000
Water absorption percent in 24 hrs/73°F	ASTM-D-570	.50
Specific gravity (minimum)	ASTM-D-792	1.50
Heat deflection at 264 psi/°F	ASTM-D-648	+430
Flame spread (radiant panel)	ASTM-E-162	15 or less
Barcol, hardness (minimum)	ASTM-D-2583	40
180°F torsional modulus, psi		
at 10 sec	ASTM-D-1043	1.25 (10^6)
at 100 sec	ASTM-D-1043	1.1 (10^6)
Hot coil ignition test		
a. Time to ignition, sec	ASTM-D-229 Method II	>50
b. Burn time, sec	ASTM-D-229 Method II	<50

3.2.3 Dissimilar materials. Intimate contact of dissimilar materials, as defined in MIL-STD-899, paragraph 5.1, which can be expected to cause galvanic corrosion, shall be avoided. When such contact cannot be avoided, an interposing insulating material shall be provided to minimize the corrosive effect.

3.3 Design and construction.

3.3.1 Understructure. The understructure shall be fabricated in accordance with Drawing No. 835039A0000. All perimeter floor support members shall be of sufficient strength to provide uniform floor support on all loading conditions and when being lifted/ rails and end frame rails shall be in the same plane as the bottom corner fittings to provide a flat perimeter supporting surface. Testing shall be in accordance with 4.7.1.

3.3.2 Panels and structure. The container frame shall be fabricated from high strength alloy steel in such a manner to provide sufficient strength for all exterior panels. No part of the frame shall extend above the roof plane. All panels shall be constructed of SMC conforming to 3.2.2 and shall be of at least 0.150 in thick-

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ness with integral reinforcing ribs of at least 0.25 in. thickness in accordance with Drawing No. 835039A0000.

3.3.3 Doors. Double doors shall be provided on one 40 in (1,011 mm) side. These doors shall be interchangeable with like doors and shall be fabricated in accordance with Drawing No. 835039A0000. The doors shall be capable of withstanding without damage; a static load of 50 lbs. (222 Newtons) acting on the edge opposite the hinge pivot; a dynamic load on the door of 13 lbs. (58 Newtons) applied to the midpoint of the locking edge acting in a horizontal plane perpendicular to the front panel of the door in both directions and when the force is increased by dropping the weight (13 lbs) a distance of 12 in. (305 mm). Testing shall be in accordance with 4.7.2.

3.3.4 Container rack, PALCON. The PALCON shall be capable of housing a PALCON container rack built in accordance with MIL-C-29447(MC). When the rack is fully erected within the PALCON, six insert containers built in accordance with MIL-C-29446(MC) shall fit within the PALCON with the PALCON doors closed. Testing shall be in accordance with 4.7.3.

3.3.5 Interchangeability. Like demountable hardware and components of the PALCON shall be interchangeable.

3.3.6 Metal fabrication. Metal use in the fabrication of equipment shall be free from kinks and sharp bends. The straightening of material shall be done by methods that will not cause injury to the metal. Shearing and punching shall be done neatly and accurately. Corners shall be square and true. All bends shall be made with controlled means in order to ensure uniformity of size and shape.

3.3.7 Facing material. Panel facings shall be fiberglass, smooth or embossed, and shall be free from bubbles, glazing, cracks, and discontinuities.

3.3.8 Bolted and riveted connectors. Bolt and rivet holes shall be accurately punched or drilled and have the burrs removed. Washers, lockwashers, or locknuts shall be provided where necessary and all bolts, nuts, and screws shall be tight. Rivet heads, when not countersunk or flattened, shall be of uniform size and transported utilizing the forklift pockets. The bottom side shape for the same diameter rivet, concentric with the rivet holes, and in full contact with the surface of the members.

3.3.9 Welding. The surface of the parts to be welded shall be free from rust, scale, paint, grease, or other foreign matter. Weld penetration shall be such as to provide transference of maximum design stress through the base metal juncture. Weld configuration and sizes shall be as specified on the applicable drawings.

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3.3.10 Castings. Castings shall be sound and free from patching, misplaced coring, warping, or defects which might render the castings unsound for use.

3.3.11 Seals. Seals shall be installed so that the fit is snug and continuous contact is maintained with mating seal faces. Mitered end junctions of vertical and horizontal seal runs shall be accurate, and gaps shall be sealed as specified on the applicable drawings.

3.4 Maintainability. The PALCON has been designed so that as much of the maintenance as possible, may be performed as far forward (preferably by the using unit), as feasible.

3.4.1 Organizational Level Maintenance. Preventive maintenance (PM) consists primarily of inspecting for cleanliness and mechanical integrity, cleaning, lubricating hinges and latches as required. PM is performed by designated Marines in the supply section at the smallest unit level. Corrective maintenance, such as replacing minor parts and repainting is performed by the unit's organic maintenance personnel. No specific MOS has been designated for second echelon maintenance; only common tools in maintenance work spaces are required.

3.4.2 Intermediate Level Maintenance. Intermediate level maintenance (3d and 4th echelons) is performed by the Maintenance Battalion, FSSG. Maintenance will include patching to maintain weatherproofness, testing the alignment of structural members and realigning as required, replacing major parts, replacing container door weatherstripping; and performing modifications as directed. Typical MOS of contact team repairmen could be 1316 (metalworker), 1341 (Engineer equipment mechanic), 2161 (machinist), and 2871 (TMDE technician) or 2874 (metrology technician).

3.4.3 Depot Level Maintenance. Depot level maintenance will be accomplished on a regional basis using organic or contractor services.

3.5 Transportability.

3.5.1 Dimensions/Load Rating. The PALCON shall be capable of being transported without damage by all forms of military vehicles capable of transporting such loads either dimensionally or by load rating. These vehicles include the 5-ton cargo truck, the Logistics Vehicle System (LVS), helicopter (either internal or external), C-130 cargo aircraft, railroad train, or ship as well as commercial carriers.

3.5.2 Air Transportability. The PALCON shall be capable of withstanding rapid decompression in accordance with MIL-STD-1791 when transported aboard cargo aircraft. The PALCON shall be designed with pressure relief devices or shall be so configured

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for air shipment as to prevent any part from becoming a projectile in the event of catastrophic loss of aircraft cabin pressure. When a pressure relief or breathing device is installed, the PALCON shall retain its weatherproofness. This test shall be done with the container rack and inserts inside, if specified (see 6.2). Testing shall be in accordance with 4.7.11.

3.6 Environmental requirements.

3.6.1 Weatherproofness. The PALCON shall withstand testing as described in paragraph 4.7.9.1 without leakage at any location.

3.6.2 Temperature-humidity. The PALCON shall be capable of functioning and withstanding exposure to temperatures of -65°F to 125°F at any possible relative humidity from 0 to 100 percent. In storage, the PALCON shall be capable of withstanding exposure to temperatures of -70°F to 160°F at any possible relative humidity from 0 to 100 percent without cracking, corroding, delamination, or peeling of paint. A humidity detection device shall be installed on each PALCON. The device should be capable of resetting itself. Testing shall be in accordance with 4.7.9.2 and 4.7.9.3.

3.6.2.1 Thermal shock. The PALCON shall be capable of withstanding sudden temperature changes up to and including from 160°F to -70°F to 160°F without degradation of physical properties. Testing shall be in accordance with 4.7.9.4.

3.6.3 Snow and ice. The PALCON shall be capable of functioning in accordance with the requirements specified herein during periods of snow and ice storms. The PALCON shall inhibit the collection of snow and ice such that its primary function is not degraded. The PALCON shall be capable of withstanding snow loads up to 20 lbs. per square foot (967.5 N/m^2) in addition to its internal load.

3.6.4 Wind. The PALCON shall be capable of withstanding wind speeds of up to 65 miles per hour (mph) (104 kilometers per hour (km/hr)) and gusts of up to 120 mph (192 km/hr). In winds in excess of 55 mph (88 km/hr), ground anchors are required.

3.6.5 Roof load. The PALCON shall withstand a uniform load of 33 lbs per square foot (1,580 N/m^2) over the entire roof of the PALCON without permanent damage and a concentrated load of 225 lbs (100 Newtons) over an area 12 in. by 24 in. (305 mm by 610 mm) without permanent damage rendering the PALCON unusable. Testing shall be in accordance with 4.7.9.5 and 4.7.9.6.

3.6.6 Sand and dust. The PALCON shall withstand the effects of sand and dust on all external moving parts without degradation or infiltration of particles inside the unit. Testing shall be in accordance with 4.7.9.7.

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3.6.7 Sunshine. The PALCON shall withstand the effects of ultraviolet radiation (sunshine) without degradation or effect on serviceability of the PALCON's components or materials during the service life, and shall withstand the test requirements of MIL-STD-810, Method 505.1. Testing shall be in accordance with 4.7.9.8.

3.6.8 Solar load. The PALCON shall withstand a solar heat load on its roof sufficient to raise the outer skin to at least 200°F without any evidence of delamination or permanent deformation. Testing shall be in accordance with 4.7.9.9.

3.6.9 Marine environment. The PALCON shall be fully serviceable when exposed to sea-salt fallout equivalent to 25 pounds/acre/year (2.8 grams/m^2/year). All fasteners, seals, or other hardware shall show no evidence of corrosion or degradation affecting serviceability when tested as required by 4.7.9.10.

3.6.10 Drop shock.

3.6.10.1 Flat drop. The PALCON shall withstand a flat drop of 12 in. (305 mm) from the ground onto a concrete surface. Testing shall be in accordance with 4.7.10.1.

3.6.10.2 Rotational drop. The PALCON shall withstand rotational drops with one edge placed on a 4 in. (102 mm) block and the opposite edge raised 12 in. (305 mm) and allowed to free fall onto a concrete surface with the 4 in. (102 mm) block acting as a pivot. The PALCON shall withstand this drop on each bottom edge. Testing shall be in accordance with 4.7.10.2.

3.7 Dimensions. The minimum internal dimensions shall be as listed in Table II. The actual external dimensions and their tolerances shall be as listed in Table III. The applicable drawings fully describe all pertinent dimensions and tolerances. Should any dimensions and tolerances result in a component or assembly which will not meet the requirements of this specification, the supplier is responsible for fabrication techniques which will prevent the occurrence of such dimensions and tolerances. Testing shall be in accordance with 4.7.7.

TABLE II. Minimum Internal Dimension

Height (in)	Width (in)	Length (in)
32 1/4	36	44 1/2

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TABLE III. Actual External Dimensions and Their Tolerances

Height (in)	Height Tolerances (in)	Width (in)	Width Tolerances (in)	Length (in)	Length Tolerances (in)
40 15/16	+0 -1/8	39 13/16	+0 -1/8	47 13/16	+0 -1/8

3.8 Weight. The tare weight of the PALCON, including all permanent accessories, shall be 378 lbs. ± 5 percent. The gross weight shall not exceed 1,250 lbs. When eight containers are coupled together to form the largest array (2 high by 2 wide by 2 long) configuration, the group of eight shall have a gross weight rating of 10,000 lbs. The actual tare weight shall be stamped on the nameplate. Testing shall be in accordance with 4.7.8.

3.9 Finish.

3.9.1 Surface preparations. All surfaces shall be prepared as specified on the applicable drawings. All aluminum alloy materials shall be protected by a chemical film conforming to MIL-C-5541 even if a final finish is not applied.

3.9.2 Priming and final finish. All surfaces to be painted shall be cleaned and treated in accordance with MIL-T-704 (Table II), type B, except that all surfaces or ferrous components shall be coated with a primer coat and topcoat in accordance with the applicable drawings.

3.10 Identification marking.

3.10.1 Identification. Each PALCON shall have the following information provided on a serial plate permanently affixed to the PALCON in accordance with MIL-STD-130:

Manufacturer:

Model:

Type:

Serial No.: (As assigned by the Contracting Activity)

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Tare Weight:

NSN:

3.10.2 Markings. Markings shall be in accordance with Drawing No. 835039A0000 and in compliance with coding and marking requirements.

3.11 Workmanship. All parts, components, and assemblies of the PALCON, including castings, forgings, molded parts, stampings, seals and sealing agents, machined surfaces, and welded parts, shall be clean and free from any defects that will reduce the capability of the PALCON to meet the requirements specified herein. Any components and assemblies which have been repaired or modified to overcome deficiencies or deficient components or assemblies which are determined to be used "as is" shall not be used unless authorized by the contracting activity. External surfaces shall be free from burrs, slag, sharp edges and corners, except where sharp edges and corners are required.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the Contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the Contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items must meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the Contractor's overall inspection system or quality program in accordance with MIL-Q-9858. The absence of any inspection requirements in this specification shall not relieve the Contractor of the responsibility of assuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling in quality conformance does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to acceptance of defective material.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.4)

b. Quality conformance inspection (see 4.5)

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c. Individual inspection (see 4.6)

4.3 Inspection conditions. Unless otherwise specified, all inspections shall be performed in accordance with the test conditions specified in 4.4, 4.5, and 4.6.

4.4 First article inspection. Prior to beginning production of the PALCON, the Contractor shall fabricate eight PALCONs and subject two of them to the tests specified in Table IV, except when four or eight PALCONs are required.

4.5 Quality conformance inspection. From each shipping lot, a single PALCON shall be chosen at random by the Government and subjected to the tests specified in Table IV. Failure of any test shall be cause for inspection of the entire lot at the Contractor's expense for similar defects. Defective PALCONs shall not be delivered without correction of the deficiency and then only with permission of the Contracting Officer.

4.6 Individual inspection. Each PALCON shall be subjected to the inspections and tests specified in Table IV. Failure to meet the requirements specified herein shall be cause for rejection. No PALCON shall be reworked and retested without the expressed permission of the contracting activity.

TABLE IV. Tests

INSPECTION	REQ PARA	TEST PARA	1ST ART	QUAL CONF	IND
Lift by fork pockets	3.3.1	4.7.1	X		
Door load	3.3.3	4.7.2	X		
Compatibility	3.3.4	4.7.3	X		
Lifting from top	- - -	4.7.4	X		
Endwall strength	- - -	4.7.5	X		
Sidewall strength	- - -	4.7.6	X		
Dimensional	3.7	4.7.7	X	X	X
Weight	3.8	4.7.8	X	X	X
Weatherproofness	3.6.1	4.7.9.1	X	X	

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INSPECTION	REQ PARA	TEST PARA	1ST ART	QUAL CONF	IND
Humidity resistance	3.6.2	4.7.9.2	X		
Temperature range	3.6.2	4.7.9.3	X		
Temperature shock	3.6.2.1	4.7.9.4	X		
Roof load, uniform	3.6.5	4.7.9.5	X		
Roof load, concentrated	3.6.5	4.7.9.6	X		
Sand and dust	3.6.6	4.7.9.7	X		
Sunshine	3.6.7	4.7.9.8	X		
Solar load	3.6.8	4.7.9.9	X		
Marine atmosphere	3.6.9	4.7.9.10	X		
Flat drop 1/	3.6.10.1	4.7.10.1	X	X	
Rotational drop 1/	3.6.10.2	4.7.10.2	X	X	
Air Transportability 1/	3.5.2	4.7.11	X		

1/ With rack and inserts if specified (see 3.5.2 and 3.6.10)

4.7 Tests.

4.7.1 Lift by fork pockets. The PALCON to be tested shall be uniformly loaded to a gross weight of 1,250 lbs. (5,560 Newtons). The PALCON shall then be supported by two horizontal bars, each 8 in. (20 mm) wide, all the way through the fork pockets on one side of the PALCON with sufficient length protruding outside the walls (doors) to attach lifting cables. The PALCON shall then be lifted clear of the ground using only the bars for a period of 5 minutes and then lowered to the ground. The procedure shall then be repeated using the set of fork pockets 90° from those already tested. Upon

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completion of this test, the PALCON shall show neither permanent deformation nor abnormality which will render it unsuitable for use and the dimensional requirements of 4.7.7 shall then be satisfied.

4.7.2 Door load. Each door of the PALCON under test shall have a vertical force of 50 lbs. (222 Newtons) applied at the edge opposite from the hinge pivot with the door open at 90°. After 30 minutes the force shall be removed and the door examined for evidence of unbonded components, damage to hardware, and improper door operation. The door shall then be opened to its maximum, a fixture shall be attached to the midpoint of the locking edge of the door which shall permit the application of 13 lbs. (58 Newtons) using free running cables to transmit the forces from the weight and 1/4 diameter steel cable. The weight of 13 lbs. (58 Newtons) shall be dropped five times from a height of 12 in. (305 mm). This shall be repeated from both sides of the door. Any evidence of deformation, damage, or improper door operation shall constitute failure of this test.

4.7.3 Compatibility. A pallet container rack built in accordance with MIL-C-29447(MC) and one or more container inserts built in accordance with MIL-C-29446(MC) shall be required for this test. Assemble the rack into the PALCON and assure compatibility of six insert locations by installing the insert container into each location.

4.7.4 Lifting from top. As a single container and as a coupled group of containers as defined in Table V, the container(s) under test shall withstand lifting by an overhead crane or sling without permanent deformation. The containers shall be tested in the maximum gross weight conditions for each of the eight load configurations as defined in Table V. The PALCON assembly shall be carefully lifted in such a way that no significant acceleration or deceleration forces are applied. The PALCON assembly shall be suspended by the lifting mechanism for 5 minutes and then carefully lowered to the ground. Upon completion of this test, each PALCON, and the assembly, shall show neither permanent deformation nor abnormality which will render it unsuitable for use and the dimensional requirements of 4.7.7 shall then be satisfied for each PALCON individually and as an assembly.

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TABLE V. Sizes and Maximum Weights of Loads

Load Configuration	Nominal Width (in)	Nominal Height (in)	Nominal Length (in)	Nominal Volume (ft^3)	Maximum Gross Weight (lb)
PALCON	40	41	48	46	1,250
2-PALCON Array (side by side)	80	41	48	91	2,500
2-PALCON Array (back-to-back)	40	41	96	91	2,500
2-PALCON Array (top-to-bottom)	40	82	48	91	2,500
4-PALCON Array (one wide)	40	82	96	182	5,000
4-PALCON Array (one long)	80	82	48	182	5,000
4-PALCON Array (one layer)	96	41	80	182	5,000
8-PALCON Array	96	82	80	364	10,000

4.7.5 Endwall strength. Place the PALCON under test on its end (side measured 48 in. long) supported to allow free deflection of the lower endwall. Uniformly load the inner surface of the lower endwall to a weight of 870 lbs. (3,870 Newtons). Upon completion of this test, the PALCON shall show neither permanent deformation nor abnormality which will render it unsuitable for use and the dimensional requirements of 4.7.7 shall then be satisfied for a single PALCON.

4.7.6 Sidewall strength. Load the PALCON under test to 870 lbs. (3,870 Newtons). Make sure the doors are securely latched. Rotate the unit carefully so that the unit is sitting with the doors down. The unit shall be supported so that the door panels are free to deflect with the doors down. Allow the weight to remain in place for 5 minutes. Rotate the unit carefully to position the weight against the panel opposite the doors. The unit shall be supported so that the loaded panel is free to deflect. Allow the weight to remain in place for 5 minutes.

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Upon completion of this test, the PALCON shall show neither permanent deformation nor abnormality which will render it unsuitable for use and the dimensional requirements of 4.7.7 shall then be satisfied for a single PALCON.

4.7.7 Dimensional. The length, width, and height shall be measured and recorded for each PALCON. During first article testing, the length, width, and height shall be measured and recorded for the eight PALCONs. The requirements of 3.7 shall be satisfied.

4.7.8 Weight. The weight of each PALCON shall be taken. The requirements of 3.8 shall be satisfied.

4.7.9 Environmental tests.

4.7.9.1 Weatherproofness. A stream of water shall be applied on all exterior joints and seams of the PALCON under test from a nozzle of 0.5 in. (12.7 mm) inside diameter at a pressure of about 1 bar (corresponding to a head of about 33 ft. (10m)) of water on the upstream side of the nozzle. The nozzle shall be held at a distance of 5 ft. (1.5m) from the container under test, and the stream shall be traversed at a speed of 4 in/s (100 mm/s). Procedures involving the use of several nozzles are acceptable provided that each joint or seam is subjected to a water loading no less than that which would be given by a single nozzle. At the conclusion of this test, the QUADCON shall be examined and show no evidence of leakage at any location.

4.7.9.2 Humidity resistance. The PALCON shall be subjected to MIL-STD-810, Method 507.1, Procedure II, excluding steps 2-4, 7, and 8. After cycling, the PALCON shall be examined and show no evidence of delamination, cracking, or deterioration of any part of the PALCON.

4.7.9.3 Temperature range test - low temperature test. The PALCON shall be cold soaked in a mechanically refrigerated cold chamber at -70°F for a minimum of 16 hours. The chamber temperature shall then be raised to -65°F. The PALCON doors will be opened and closed. Any malfunction shall be recorded and reported. The PALCON shall then be visually examined both internally and externally for material degradation. There shall be no damage to seals or other components and all hardware shall operate as specified during and upon completion of this test.

4.7.9.4 Temperature shock. A representative PALCON panel specimen measuring 22 in. by 37.38 in. (559 mm by 949 mm) shall be subjected to MIL-STD-810, Method 503.1. Upon completion of this test, there shall be no separation, cracks, delamination, or degradation of physical properties. Superficial cracks in the glazing which will not affect serviceability of the material are acceptable. The government may require testing to confirm that the physical properties are not affected.

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4.7.5 Roof load, uniform. The PALCON under test shall be placed on the ground and leveled. The entire roof of the PALCON shall then be uniformly loaded with a weight of 450 lbs. (33 lbs. per square foot (1,580 N/m) and allowed to remain loaded for a period of 5 minutes. Upon completion of this test, the PALCON shall show neither permanent deformation nor abnormality which will render it unsuitable for use and the dimensional requirements of 4.7.7 shall then be satisfied.

4.7.9.6 Roof load, concentrated. A load of 225 lbs. (100 Newtons) shall be uniformly distributed over an area of 12 in. by 24 in. (305 mm by 610 mm) located at the weakest (nonsupported) area of the roof of the PALCON. This load shall be allowed to remain there for a period of 5 minutes. Upon completion of this test, the PALCON shall show neither permanent deformation nor abnormality which will render it unsuitable for use and the dimensional requirements of 4.7.7 shall then be satisfied.

4.7.9.7 Sand and dust. The PALCON shall be subjected to MIL-STD-810, Method 510.1, Procedure I. There shall be no damage to seals or other components and all hardware shall operate as specified during and upon completion of this test.

4.7.9.8 Sunshine. Representative samples of all gaskets and seals, as well as a 12 in. by 12 in. (305 mm by 305 mm) panel sample, shall be subjected to MIL-STD-810, Method 505.1, Procedure I. Upon completion of this test, the samples shall be visually examined and any evidence of degradation of physical properties shall constitute failure of this test.

4.7.9.9 Solar load. The PALCON shall be subjected to a simulated solar load sufficient to raise the outer skin temperature to a minimum of 200°F uniformly over the fixed roof. The uniform solar load temperature should be gradually attained within 4 hours and shall be maintained for an additional 4 hours. As a minimum, one thermocouple per 10 square feet (one square meter) shall be uniformly distributed on the entire area of the roof. All of the thermocouples should read 200°F ± 15°F throughout the 4 hour period that the solar load shall be maintained. Upon completion of this test, the roof panels shall be examined and any evidence of delamination or deformation shall constitute failure of this test.

4.7.9.10 Marine atmosphere. One representative sample of each fastener, seal, panel, and hinge which will be exposed to the atmosphere in the operation and storage modes shall be finished in accordance with the applicable drawings and subjected to tests as described in MIL-STD-810, Method 509.1, except that the salt solution shall have a concentration of 10 percent and the exposure period shall be 96 hours. Upon completion of this test, the test items shall show no evidence of delamination, cracking, corrosion, or deterioration.

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4.7.10 Shock tests.

4.7.10.1 Flat drop. The PALCON shall be uniformly loaded to the rated gross weight of 1,250 lbs. (5,560 Newtons) and lifted 12 in. (305 mm) above a flat concrete surface. The distance between the concrete surface and each corner of the PALCON under test shall not vary more than 0.5 in. (12.7 mm) at the time of the drop. The PALCON shall then be allowed to free fall to the concrete surface. This test shall be done with the container rack and inserts inside, if specified in the contract (see 3.6.10). Upon completion of this test, the PALCON shall show neither permanent deformation nor abnormality which will render it unsuitable for use and the dimensional requirements of 4.7.7 shall then be satisfied.

4.7.10.2 Rotational drop. The PALCON shall be uniformly loaded to the rated gross weight of 1,250 lbs. (5,560 Newtons) with one edge supported 4 in. (102 mm) above a flat concrete surface. The opposite edge shall be lifted a distance of 12 in. (305 mm) above the concrete surface and allowed to free fall onto the surface. This procedure shall be repeated until each edge has been allowed to free fall. This test shall be done with the container rack and inserts inside, if specified in the contract (see 3.6.10). Upon completion of this test, the PALCON shall show neither permanent deformation nor abnormality which will render it unsuitable for use and the dimensional requirements of 4.7.7 shall be satisfied.

4.7.11 Air Transportability. The PALCON shall be subjected to MIL-STD-1791, paragraph 5.2.5.2, Rapid Decompression Verification and shall meet the requirements for certification in accordance with MIL-STD 209.

4.8 Test reports. The results of all testing shall be reported and written in compliance with MIL-STD-831 fully describing and illustrating the test procedures and test results.

5. PACKAGING

5.1 Preservation. The PALCON requires no additional preservation.

5.2 Packing. Conventional methods of blocking, bracing, and strapping are to be employed during shipment by rail, road, and air as described in MIL-STD-1186. Removable markings shall be affixed in accordance with MIL-STD-129 to show cube, weight, and destination specified by the contracting activity.

5.3 Marking. In addition to the marking required by 3.10, each PALCON shall be marked for shipment in accordance with the requirements of MIL-STD-129. A removable data plate shall be installed to show cube, weight, and destination specified by the contracting activity.

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6. NOTES

6.1 Intended use. The PALCON covered by this specification is one of a family of containers intended for use by the Fleet Marine Forces to house and support the storage and movement of organizational property and consumable supplies in the field and in garrison.

6.2 Ordering data.

6.2.1 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of this specification.

b. When first article testing is required (see 3.1).

c. When drop shock testing with the container rack and inserts inside is required (see 3.6.10).

d. When a quality control system is required (see 4.1.1.).

6.2.2 Data requirements. When this specificaiton is used in an acquisition and data are required to be delivered, the data requirements identified below shall be developed as specified by an approved Data Item Description (DD Form 1664) and delivered in accordance with the approved Contract Data Requirments List (CDRL), incorporated into the contract. When the provisions of DOD FAR Supplement, Part 27, Sub-Part 27.410-6 (DD Form 1423) are invoked and the DD Form 1423 is not used, the data specified below shall be delivered by the contractor in accordance with the contract or purchase order requirements. Deliverable data required by this specification are cited in the following paragraphs.

Para No.	Data Requirement Title	Applicable DID No.	Option

(Data item descriptions related to this specification and identified in section 6 will be approved and listed as such in DOD 5000.19L, Vol. II, AMSDL. Copies of data item descriptions required by the Contractor in connection with specific acquisition functions should be obtained from the Naval Publications and Forms Center or as directed by the Contracting Officer.)

6.3 First article. First article test items shall be eight sample items manufactured using the same procedures and tooling proposed for production. The Contractor shall submit a first article test plan and receive Government approval prior to testing. The first articles shall be examined and tested for approval at the Contractor's plant or at an independent laboratory acceptable to the Government. First article tests shall be witnessed by a

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Government representative of the acquiring activity. After completion of first article testing, the Contractor shall submit a separate test report. The Government reserves the right to waive the requirement for first article testing for items that have previously been tested and accepted by the Government.

6.4 Subject term (key word) listing. The following subject term (key word) listing is furnished for retrieval searches.

Container

Container, Pallet

PALCON

Preparing activity
Navy-MC
Project No. 8145-0034

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-C-29444(MC)	Container, Pallet (Palcon)

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ USER
	☐ MANUFACTURER
	☐ OTHER *(Specify)* _____

5. PROBLEM AREAS

a. Paragraph Number and Wording

b. Recommended Wording

c. Reason/Rationale for Recommendation

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

DD FORM 82 MAR 1426 PREVIOUS EDITION IS OBSOLETE

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